Li Shen, PhD

Worcester, MA | lshen@clarku.edu | (716) 817-3989 | linkedin.com/in/lily-shen

SUMMARY

Ph.D.-trained structural and applied microeconomist with research in quantitative spatial economics and trade. Builds calibrated equilibrium models to study regional adjustment and counterfactual shocks, with additional experience in panel econometrics, causal-inference discipline, and pricing-demand modeling. Strong programming skills in Python, SQL, Stata, MATLAB, R, and SAS.

EDUCATION

Clark University — **Worcester, MA**
Ph.D. in Economics — *2026*

The State University of New York at Buffalo — **Buffalo, NY**
M.A. in Economics — *May 2019*

Valparaiso University — **Valparaiso, IN**
M.S. in International Economics and Finance — *May 2018*

Chongqing University — **Chongqing, China**
B.Mgmt. in Accounting — *June 2014*

TECHNICAL SKILLS

- **Structural economics:** Quantitative spatial equilibrium, quantitative trade, multi-region input-output networks, calibration, labor mobility, structural inversion, and counterfactual simulation.
- **Applied econometrics:** Panel-data methods, difference-in-differences, shift-share IV, staggered adoption designs, event studies, gravity models, and time-series forecasting (ARIMA).
- **Programming:** Python, SQL, Stata, MATLAB, R, SAS.
- **Data and communication:** GeoPandas, ArcGIS, Tableau, ggplot2, LaTeX; experience presenting technical results to academic and nontechnical audiences.

SELECTED RESEARCH AND PROFESSIONAL EXPERIENCE

Structural Spatial Economics and Quantitative Trade — Trade Shock Propagation (Job Market Paper) 2023–Present

- Co-developed a multi-sector, 31-province structural spatial-equilibrium model with inter-provincial input-output networks, non-tradable production, and labor mobility. My contributions included quantitative model development, structural parameter recovery, calibration-data work, and counterfactual analysis using China's 2017 MRIO, ICIO, and Customs data.
- Established that local trade multipliers are equilibrium objects shaped by domestic production structure rather than universal parameters, and identified the own-trade share—not raw export exposure—as the structural organizing statistic for multiplier heterogeneity.
- Extended the quantitative spatial economics framework from national welfare analysis to sub-national shock amplification; counterfactual simulations yield model-implied GDP multipliers ranging from 0.30 to 1.02 across provinces, with inland economies amplifying external shocks through non-tradable service demand.

High-Speed Rail and Market Integration 2021–Present

- Built the largest daily-frequency panel of Chinese vegetable wholesale prices to date: 3.99 million city-pair-month observations covering ten commodities across 117 prefecture-level cities (2014–2022), merged with AutoNavi smartphone-based intercity passenger mobility indices.
- Leveraged two institutional features—China's Green Passage highway-toll exemption (fixing variable freight costs) and prohibitive HSR freight tariffs for bulk produce—to study a setting in which differences in HSR connectivity reflect variation in passenger rather than freight costs.
- Documented that HSR-connected city pairs exhibit 2–5% lower intercity price dispersion; passenger-mobility controls absorb the direct HSR coefficient, and cross-product heterogeneity (larger effects for bulky, low-value commodities) argues against a freight substitution channel.

Applied Econometric Research and Forecasting 2017–2019

- Forecasted international passenger demand with seasonal ARIMA models in SAS and estimated GDP-demand elasticities for capacity planning.
- Modeled determinants of corporate revenue using regression analysis on time-series and accounting data.

Financial Auditor, Datong Jiaye Property Management Co., Ltd. 2014–2016

- Conducted internal audits, reconciled inventory records, and prepared general-ledger journal entries.

APPLIED PROJECTS

Pricing and Promotion Decision Engine — Demand Estimation, Optimization, and Deployment 2025–Present

Live app: pricing-promotion-decision-engine.streamlit.app *Code:* github.com/Li-Shen-Clark/pricing-promotion-decision-engine

- Built an end-to-end pricing decision-support application on 6.6M rows of Dominick's Finer Foods scanner data; cleaned to a 4.65M-row panel covering 486 UPCs, 93 stores, and 366 weeks.
- Estimated a log-log fixed-effects demand model with Duan smearing retransformation to forecast demand responses to price and promotion changes.
- Implemented counterfactual price and promotion simulations plus constrained profit optimization under cost, inventory, and margin constraints.
- Designed A/B validation plans with power analysis for raise-and-test pricing candidates; deployed as a Streamlit app with modular Python code, reproducible requirements, GitHub auto-deployment, and pytest tests.

Vegetable Price Intelligence — Forecasting, Risk, and Model Governance 2026–Present

Portfolio case study: Case Study *Code:* GitHub

- Built an economics-informed price intelligence system from 8.68M historical wholesale-market records across 117 cities and 30 vegetables, producing 7.36M auditable city-day facts with versioned mapping, quality flags, and coverage tiers.
- Developed rolling-origin 7/14/28-day forecasts and validation-frozen 14-day spike-risk signals; on a sealed final test, reduced WAPE from 18.72% to 17.54% and routed weak product-horizon groups to explicit baseline fallbacks.

China Population Density Atlas — Interactive Geospatial Visualization 2022–Present

Live app: pop-density-spatial.streamlit.app

- Processed 19 years of 5 km-resolution population-density rasters (2002–2020) covering China into a harmonized annual panel for grid-cell and province-level analysis.
- Built the underlying density, agglomeration, and urban-corridor metrics and deployed an interactive Streamlit application for map-based exploration.

MACHINE LEARNING COMPETITIONS

Home Credit — Credit Risk Model Stability (Kaggle Silver Medal) Feb–May 2024

Stack: Python, Polars, scikit-learn, LightGBM, CatBoost

- Co-developed a Polars lazy-execution ETL pipeline for multi-depth relational data; designed aggregation features (max, last, mean) capturing borrower behavioral trends; optimized directly for the headline Gini Stability metric to prevent performance drift over time.
- Implemented correlation-based feature grouping (Pearson > 0.8) for multicollinearity control and type-downcasting memory optimization (>50% reduction).

ICR — Identifying Age-Related Conditions (Kaggle Bronze Medal) May–Aug 2023

Stack: Python, TabPFN, XGBoost, imbalanced-learn

- Designed a stratified resampling strategy using auxiliary Greek metadata (RandomOverSampler on Alpha-defined latent subgroups) to preserve population structure under severe class imbalance.
- Co-developed a voting ensemble combining parametric (XGBoost) and non-parametric (TabPFN) models under a custom Balanced Log Loss metric; calibrated decision thresholds to minimize false negatives.

TEACHING AND HONORS

- **Teaching Assistant, Clark University** (2020–2025): Graduate finance — *Investments*, *Computational Finance* (Monte Carlo methods, stochastic processes, Black-Scholes pricing), *Case Studies in Derivatives*, and *Investment Strategy*. Undergraduate economics — *Macroeconomic Theory* (Solow growth model, IS-LM and AS-AD frameworks, Phillips-curve dynamics, monetary and fiscal policy) and *Economics and the World Economy* (comparative advantage, exchange-rate regimes, international capital flows, economic development).
- **Honors:** Sheftel Award (Research Excellence, twice, 2025), Veendorp Best Field Paper Award (2022), Doctoral Fellowship (2019–2026).